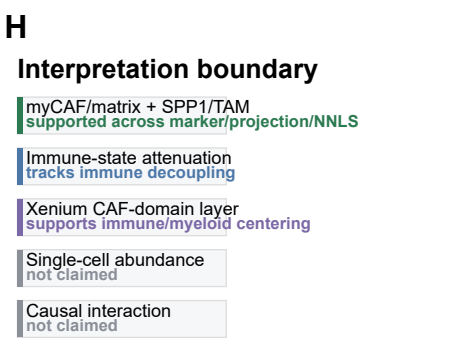
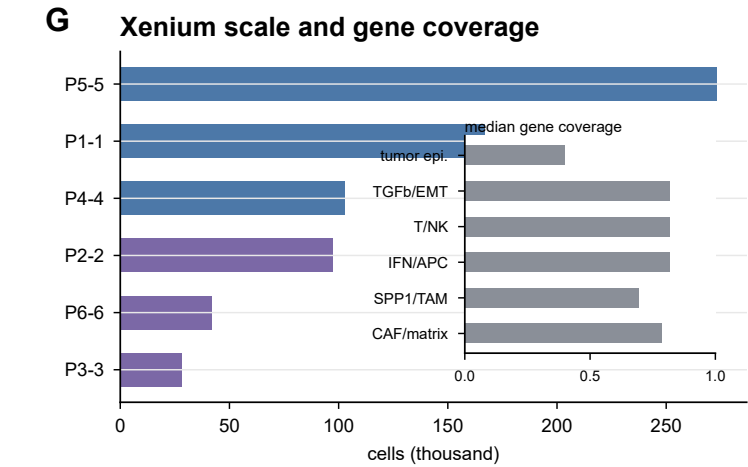
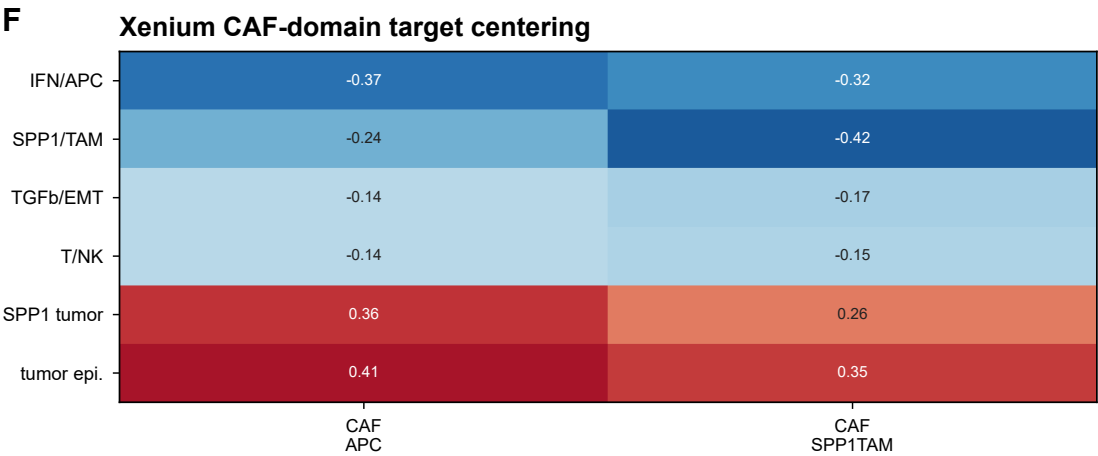
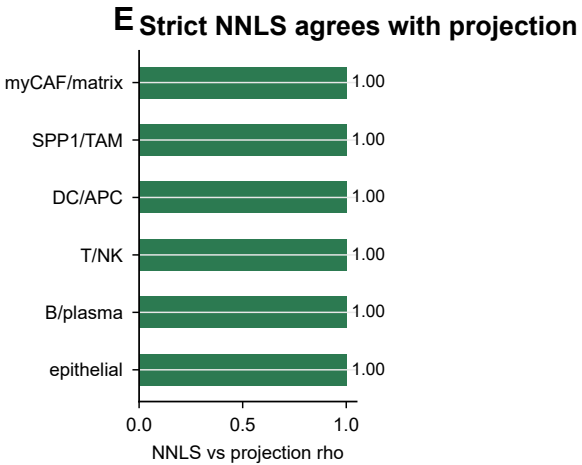
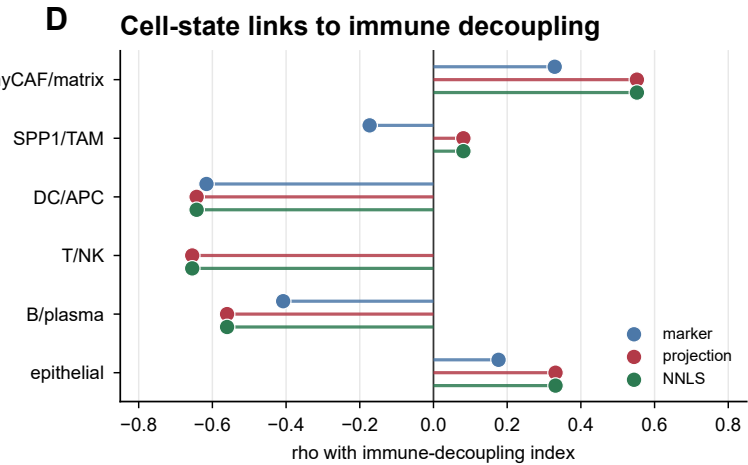
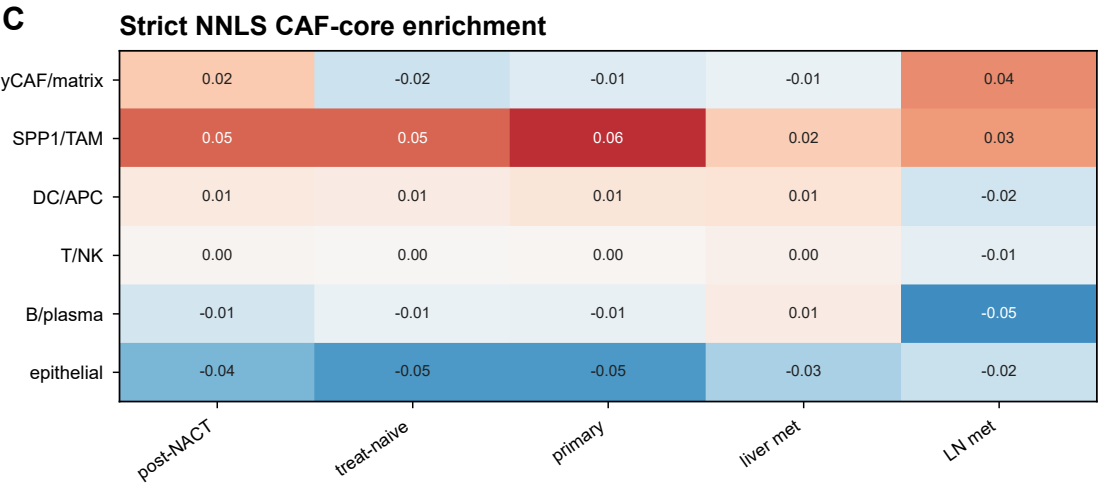
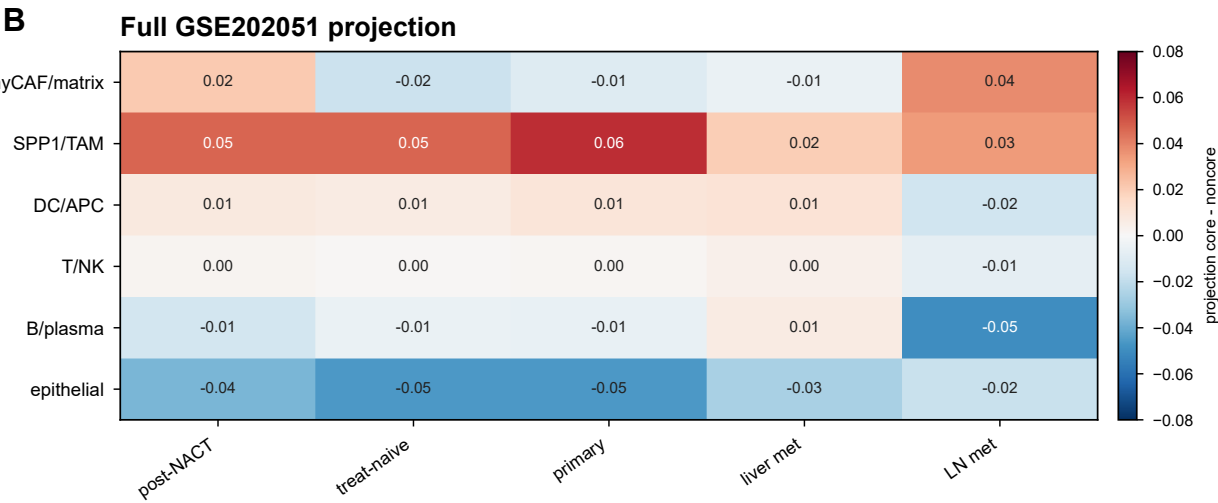
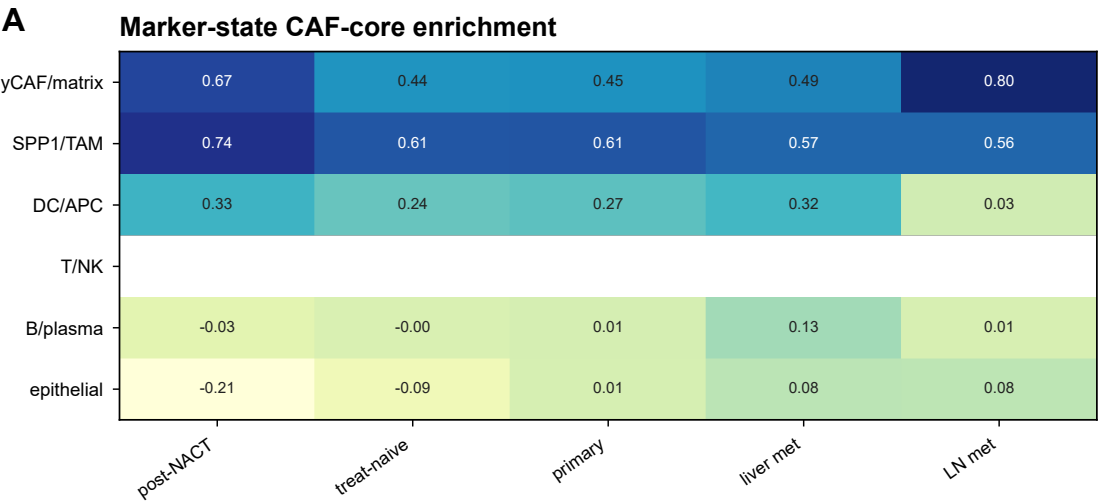


Cell-state interpretation and multi-resolution validation



Interpretation: marker-state enrichment, full GSE202051 projection, strict NNLS sensitivity and Xenium cell-resolution anchors converge on a cellular interpretation of the CAF-myeloid niche: myCAF/matrix and SPP1/TAM-rich states align with CAF cores, while immune-state coupling is attenuated in immune-decoupled contexts.

Boundary: these are expression-derived and reference-dependent support layers. They should not be described as immunostaining, image segmentation, direct abundance measurement or causal cell-cell interaction evidence.